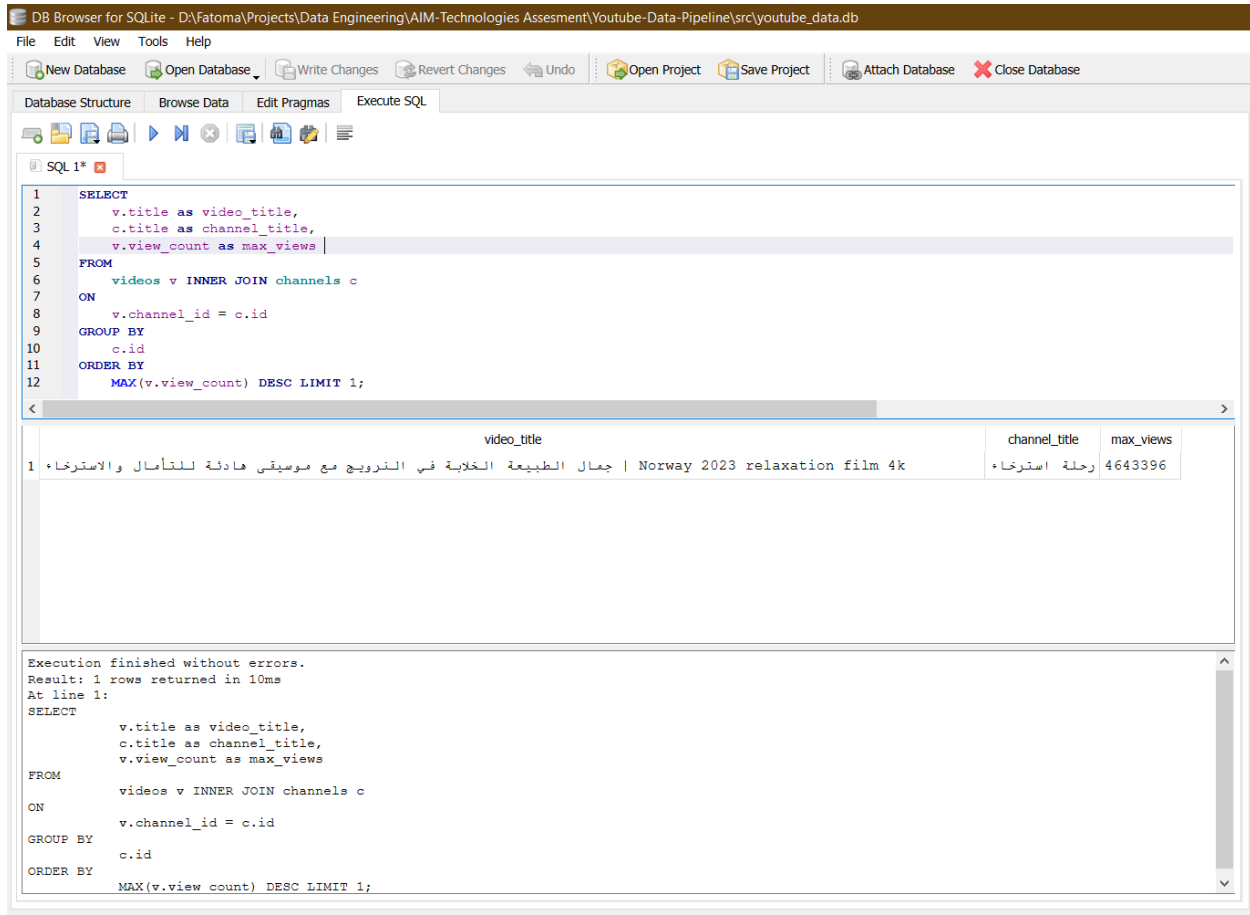


Part 4 — Analytical Queries

Top video View Count



The screenshot shows the DB Browser for SQLite interface. The title bar indicates the database path: D:\Fatoma\Projects\Data Engineering\AIM-Technologies Assesment\Youtube-Data-Pipeline\src\youtube_data.db. The menu bar includes File, Edit, View, Tools, and Help. The toolbar contains icons for New Database, Open Database, Write Changes, Revert Changes, Undo, Open Project, Save Project, Attach Database, and Close Database. The main window has tabs for Database Structure, Browse Data, Edit Pragmas, and Execute SQL. The SQL editor shows a query to find the video with the highest view count. The results pane displays one row of data.

```
1 SELECT
2   v.title as video_title,
3   c.title as channel_title,
4   v.view_count as max_views
5 FROM
6   videos v INNER JOIN channels c
7 ON
8   v.channel_id = c.id
9 GROUP BY
10  c.id
11 ORDER BY
12  MAX(v.view_count) DESC LIMIT 1;
```

	video_title	channel_title	max_views
1	جمال الطبيعة الخلابة في النرويج مع موسيقى هادئة للتأمل والاسترخاء Norway 2023 relaxation film 4k	رحلة استرخاء	4643396

Execution finished without errors.
Result: 1 rows returned in 10ms
At line 1:
SELECT
 v.title as video_title,
 c.title as channel_title,
 v.view_count as max_views
FROM
 videos v INNER JOIN channels c
ON
 v.channel_id = c.id
GROUP BY
 c.id
ORDER BY
 MAX(v.view count) DESC LIMIT 1;

Top video Comment Count

DB Browser for SQLite - D:\Fatoma\Projects\Data Engineering\AIM-Technologies Assesment\Youtube-Data-Pipeline\src\youtube_data.db

File Edit View Tools Help

New Database Open Database Write Changes Revert Changes Undo Open Project Save Project Attach Database Close Database

Database Structure Browse Data Edit Pragmas Execute SQL

SQL 1*

```
1 SELECT
2     v.title as video_title,
3     c.title as channel_title,
4     v.comment_count as max_views
5 FROM
6     videos v INNER JOIN channels c
7 ON
8     v.channel_id = c.id
9 GROUP BY
10    c.id
11 ORDER BY
12    MAX(v.comment_count) DESC LIMIT 1;
```

	video_title	channel_title	max_views
1	جمال الطبيعة الخلابة في النرويج مع موسيقى هادئة للتأمل والاسترخاء Norway 2023 relaxation film 4k	رحلة استرخاء	767

Execution finished without errors.
Result: 1 rows returned in 7ms
At line 1:
SELECT
v.title as video_title,
c.title as channel_title,
v.comment_count as max_views
FROM
videos v INNER JOIN channels c
ON
v.channel_id = c.id
GROUP BY
c.id
ORDER BY
MAX(v.comment count) DESC LIMIT 1;

Average views per channel



SQL 1*

```

1  SELECT
2      c.title as channel_title,
3      AVG(v.view_count) as avg_views
4  FROM
5      videos v INNER JOIN channels c
6  ON
7      v.channel_id = c.id
8  GROUP BY
9      c.title
    
```

	channel_title	avg_views
1	Arabs Data Talks	1792.03333333333
2	Data Engineering	6710.00763358779
3	رحلة استرخاء	230645.79113924

Execution finished without errors.

Result: 3 rows returned in 11ms

At line 1:

```

SELECT
    c.title as channel_title,
    AVG(v.view_count) as avg_views
FROM
    videos v INNER JOIN channels c
ON
    v.channel_id = c.id
GROUP BY
    c.title
    
```

Videos published last 30 days now 2/7

DB Browser for SQLite - D:\Fatoma\Projects\Data Engineering\AIM-Technologies Assesment\Youtube-Data-Pipeline\src\youtube_data.db

File Edit View Tools Help

New Database Open Database Write Changes Revert Changes Undo Open Project Save Project Attach Database Close Database

Database Structure Browse Data Edit Pragma Execute SQL

SQL 1*

```
1 SELECT
2   c.title as channel_title ,
3   v.title as video_title ,
4   v.published_at,
5   v.view_count
6 FROM
7   videos v INNER JOIN channels c
8 ON
9   c.id = v.channel_id AND
10  v.published_at >= datetime('now', '-30 days')
11 ORDER BY
12  v.published_at DESC;
```

	channel_title	video_title	published_at	view_count
1	رحلة استرخاء - رحلة الجبال	سويسرا التي تشبه الحلم - طبيعة ريفية ونهر وبحيرة بين الجبال Appenzell 4K	2026-06-27 14:00:21.000000	106823
2	رحلة استرخاء - رحلة	طبيعة سويسرا بين قرى حجرية توقف فيها الزمن Swiss Stone Villages 4K Nature Sounds	2026-06-17 15:56:51.000000	149051

Execution finished without errors.
Result: 2 rows returned in 16ms
At line 1:
SELECT
c.title as channel_title ,
v.title as video_title ,
v.published_at,
v.view_count
FROM
videos v INNER JOIN channels c
ON
c.id = v.channel_id AND
v.published_at >= datetime('now', '-30 days')
ORDER BY
v.published_at DESC;